

Data & Decisions: Section 1

Samples and Surveys

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- **GSI notes** folder on *bCourses*
- **Emails:** please include *XMBA200S* in the object
 - ▷ sfarina@berkeley.edu
 - Preferred to messages on bCourses
- **Weekly OHs**
 - ▷ **Weekly Tuesdays 5pm PST**
 - ▷ **always same zoom link**
 - Your doubts on concepts/exercises
 - If you cannot make it, we'll find an alternative slot

- **Discussion Sections**

- ▷ 10/10; 10/31 - 10:30am -12:00pm
- ▷ same zoom link
- Sections are a resource for you: review of concepts + exercises
- Please raise your hand/use your voice (no chat)
- Sections are recorded: recordings in *bCourses*

- I am **your learning companion**
- I only grade tests. OHs and Sections are **safe space!!**
- I want you *not* to be stressed bc of this course
- I want you to have a good grade + learn sth **useful and impactful** for your career

Questions?

1. Review Ch. 13

Sampling - bias

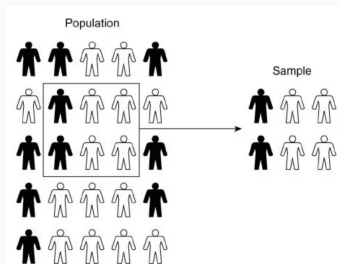
2. Review Ch. 14

Sampling - variation

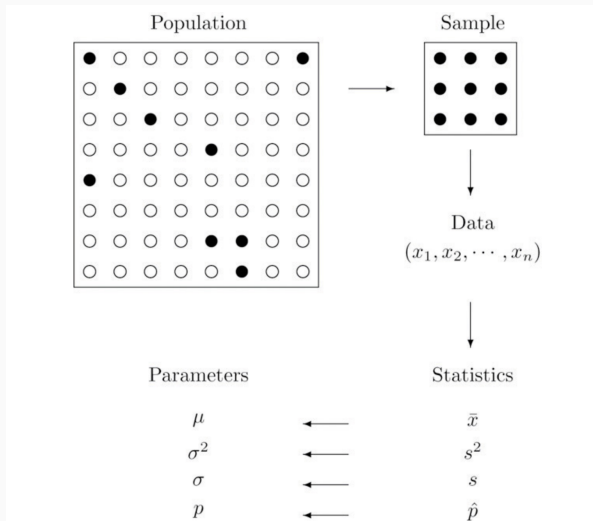
Review Ch. 13

Sampling

- We want to answer questions about the **population**
- But collecting data from the entire population is infeasible or impractical $\Rightarrow$ we use a **sample**
- By convention: **N** = size of the population
n = size of the sample



Statistics in one picture



Definitions

- **Population:** All individuals/events of interest. It is the entire group that you want to draw conclusions about!
 - **Sample:** A subset of the population
-
- **Census:** Collecting data on the **population**
 - **Survey:** Collecting data on a **sample**
-
- **Parameter:** A characteristic of the **population**
 - **Statistic:** A characteristic of a **sample**

Block 1: we focus on SAMPLING
Data you want vs Data you have

- ✓ A sample is **representative** if it *reflects* a snapshot of the population
- ✗ A sample is **biased** if it *distorts* the population

How to achieve a **representative sample** $\Rightarrow$ **randomization**

Randomization: exercise

Name the sampling method used in each design.



<https://forms.gle/8uVaHjNTBabzxRrc9>

Randomization: exercise

Name the sampling method used in each design.

1. A pollster splits 8,000 registered voters into four age groups and takes an SRS of 50 voters from each group.
2. A principal numbers all 1,450 students in the school directory and uses a random number generator to draw 100 different numbers.
3. A district numbers its 60 bus routes, picks 6 routes at random, and surveys every student who rides those 6 routes.

Randomization

Simple random sample



Systematic sample



Stratified sample



Cluster sample



Bias in Survey Design: exercise

A city wants to estimate the proportion of its adults who have a primary care doctor.

1. Design A dials 600 randomly chosen listings from the residential landline directory and 540 people answer.
2. Design B mails a survey to 800 addresses drawn at random from the city address list and 168 are returned.

→ Name the main bias in each design and give a likely direction.

- **Sampling Frame Bias:**
- **Voluntary Response Bias:**

- **Sampling Frame Bias**: some members of a population are systematically more likely to be selected in a sample than others
- **Voluntary Response Bias**: a group of individuals are invited to respond, but only those who respond are counted in the sample

Bias in Survey Design: exercise

- **Design A: Sampling frame bias** Adults in households with no landline have no chance of being selected at all.
→ Design A likely **overstates** the true proportion.
- **Design B: Voluntary response bias**
→ People who take the time to return a health survey are plausibly those already connected to regular care, so Design B likely **overstates** the proportion as well.

Which one *easier* to fix?

Bias in Survey Design: exercise

- Design A's flaw is in the **sampling frame**.
- Design B's frame is ok, only the **response step failed**.
 - Design B easier to fix: follow-up mailings, a second contact, or in-person visits to a random subset of the non-responders

Note that the 90% response rate does not make Design A the better study.

→ A high response rate within a **frame that misses part of the population** still leaves an estimate that is systematically wrong.

Many investors have money in hedge funds. To learn about the fees charged by these funds, an analyst develops a comprehensive list of all active hedge funds and selects a SRS for her analysis.

- **Survivor Bias:** units are in the sample because they have survived an "elimination" process. Units that have survived are likely to be very different from those who did not

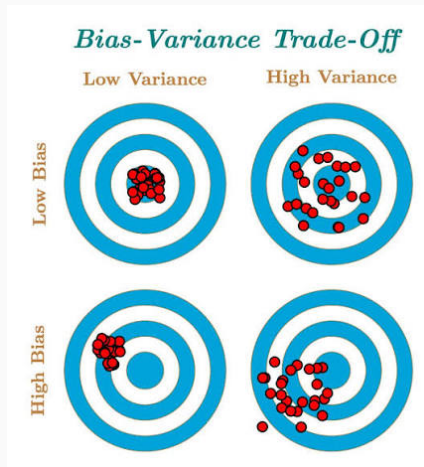
Now we have a representative sample
So what?

Review Ch. 14

You need to make decisions observing limited data (sample). When a sample at hand, you should think about two things:

- **Bias:** your answers/decisions may systematically be wrong
⇒ you need a representative sample
- **Sampling variation:** your answers/decision won't be exactly right, but statistics can tell us how close we are to being right
⇒ this is why statistical inference

Sample: bias vs variance



- We have a population with **mean** μ (known or unknown)
- We randomly take a sample of n observations and calculate the **sample mean** $\bar{X}$
- ▶ **The sample mean has a sampling distribution** **Why?**
- ▶ For **sufficiently large n** , the **sample mean is approximately normally distributed** (CLT)

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

- What does sufficiently large n means? That is, when can I apply CLT?
 - ▷ Sample size $n > 10|K_4|$ (your book)
 - ▷ Sample size $n > 30$ (other books)
- ★ If population is normally distributed, you don't even need a large enough sample size

This property on **sample mean distribution** is really nice since:

- We can take any weird looking dataset, and work with the sample mean
- Normal distributions are easy to work with
- Once we know $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$, we can **normalize** it so that we can use the **z-table**

$$Z = \frac{\text{Var-mean}}{sd}$$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

CLT Simulator link

Let's stop and think about what the distribution of the sample mean tells us

- The standard error is proportional to σ
 - When the underlying data is more variable, the sample mean is more variable
- The standard error is inversely proportional to $\sqrt{n}$
 - The larger the sample size, the smaller the sampling variation
- The standard error does not depend on the size of the population
 - Larger population do not require larger sample

Now let's start making decisions

- Both rely on the **sampling distribution** of a statistic (NO parameter!)
- **Control limits**
 - We know the population parameters
 - They determine the range in which $\bar{X}$ should be found
 - We extract a sample and we have a rule to determine whether our process is working properly (i.e. the sample is extracted from the right population)
- **Confidence Intervals**
 - We don't know the population parameters
 - We extract a sample to have a range of plausible values for the unknown parameter

Control Limits

- We know the parameter of a process
- We want to know whether the process is functioning correctly (i.e., whether to stop it or continue it)
- We extract a sample to make such a decision
- How to distinguish sampling variation vs out-of-control variation?

Control limits: Setting

- The **population mean** μ and s.d. σ are **known**.
- We take a sample of the outcomes of the process
- We compute the **sample mean** $\bar{X}$
- We define the **control limit** as a range $[\mu - L; \mu + L]$
- ✓ If $\bar{X}$ is close enough to μ (i.e. $\mu - L \leq \bar{X} \leq \mu + L$) then, production continues
- ✗ If $\bar{X}$ is far from μ (i.e. $\bar{X} < \mu - L$ or $\bar{X} > \mu + L$) then, production stops

Control Limits

- **Type I:** The mistake of taking action when no action is needed
- **Type II:** The mistake of failing to take action when needed.

		The Truth (Based on Entire Population)	
		Nothing Is There (H_0 Is True)	Something Is There (H_0 Is False)
Your Conclusion (Based on Your Sample)	I Don't See Anything (Nonsignificant)	Right!	Wrong (Type II Error)
	I See Something (Significant)	Wrong (Type I Error)	Right!

Control Limits

- **Type I:** The mistake of taking action when no action is needed
 - Stop production when NOT needed (i.e. process is working properly)
- **Type II:** The mistake of failing to take action when needed
 - DO NOT stop production when needed (i.e. process is not working properly)

		The Truth (Based on Entire Population)	
		Nothing Is There (H_0 Is True)	Something Is There (H_0 Is False)
Your Conclusion (Based on Your Sample)	I Don't See Anything (Nonsignificant)	Right!	Wrong (Type II Error)
	I See Something (Significant)	Wrong (Type I Error)	Right!

Our rule is a range $[\mu - L; \mu + L]$. **How to determine it?**

- Wide control limits reduce the chance for a Type I error
- Narrow control limits reduce the chance for a Type II error
- Trade-off: cannot simultaneously reduce the chances of both
- Convention: **focus on Type I error** whose probability is typically set at 5% or 1%

Types of problems can be of 2 categories

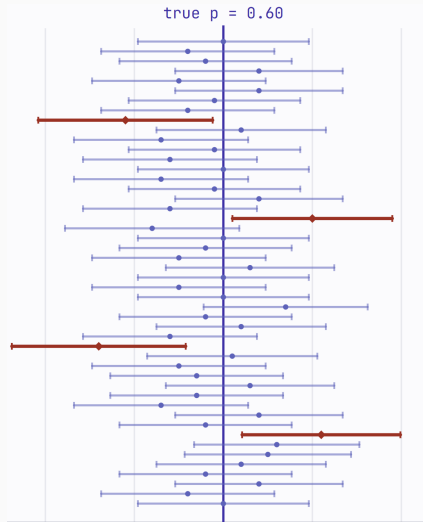
1. Given the rule (i.e. the control limit range), **compute $\Pr(\text{Type I error})$**
2. Given $\Pr(\text{Type I error})$, **find the rule (i.e. the control limit range)**
 - See the exercise on HALT uploaded on *bCourses* in subfolder "GSI notes"

Confidence Intervals

Confidence Intervals

- **Confidence Intervals** provides a range of **plausible values** for an **unknown parameter** based on your sample
- The length of the interval conveys the precision of the estimate
- **Short confidence intervals** $\implies$ **precise estimates**
- We have seen CIs for 2 parameters: **mean (μ)** and **proportion (p)**

CIs: graphic intuition



Simulator link

CIs: Formula

	Proportion	Mean
Population Parameter	p	μ
Sample Statistic	$\hat{p}$	$\bar{X}$
Standard error (SE)	$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$\frac{s}{\sqrt{n}}$
Critical Value	$z_{\alpha/2}$	$t_{\alpha/2, n-1}$
Confidence Interval (CI)	$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$\bar{X} \pm t_{\alpha/2, n-1} \times \frac{s}{\sqrt{n}}$

When **sample size conditions** hold

- For proportion: $n\hat{p} > 10$ and $n(1 - \hat{p}) > 10$
- For mean: $n > 30$ or $n > 10$ $|K_4|$

⇒ **You can use z critical value** so that formulas become

Variable Type	Statistic	CI Formula
Proportion/Binary	$\hat{p}$	$\hat{p} \pm z_{\alpha/2} * \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
Continuous	$\bar{X}$	$\bar{X} \pm z_{\alpha/2} * \frac{s}{\sqrt{n}}$

CIs: Interpretation

If I tell you: The $100(1 - \alpha)\%$ confidence interval for the **population parameter** calculated from the sample is $[a;b]$.

What does that mean?

- ⇒ I'm $100(1 - \alpha)\%$ **confident** that the value of the **true population parameter** is between $[a;b]$.
- ⇒ If we were to take 100 different samples and compute CIs for each sample, then approximately $100(1 - \alpha)$ of the 100 confidence intervals will contain the **true population parameter**
- * *Will only accept these two interpretations*

Other questions you may have about the interpretation

1. Could it happen that the **population parameter** is **outside** the interval $[a;b]$?
 - ▶ Yes, but this is NOT compatible with our sample at $100(1 - \alpha)\%$ level of confidence
2. Could I be **more confident** ? Say from 95% to 99%?
 - ▶ Yes, but CIs will become wider, hence less precise estimate of the population parameter
 - ▶ [usually, not interesting and not done]

3. Could I be **more precise** at the *same level of confidence*?

- ▶ Yes, a larger sample size will reduce the estimated standard error of the statistic, hence a narrower CI
- ▶ [a larger sample is always better]

1. You don't know the population parameter (μ)
2. You extract a sample and compute the sample statistic ($\bar{X}$)
3. You compute the standard error of the sample statistic ($se(\bar{X})$)
4. You are given the confidence level $100(1 - \alpha)\%$
5. You apply the formula
6. You conclude by **copying** the interpretation we have seen together

How large a sample do we need?

- I want to say something about the population parameter at 95% level of confidence.
- ▷ Extract a sample and compute a CI
- As before + I want to say sth within a given range (i.e. precision)
- ▷ Before CI, compute the needed sample size through the margin of error formula

Margin of Error

You don't care about t or z: you just have 2 as a value

- For sample mean: $ME = 2 \frac{s}{\sqrt{n}}$

▷ $n = \frac{4s^2}{ME^2}$

▷ You need pilot sample to have s

- For proportion: $ME = 2 \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

▷ $n = \frac{4(1/2)^2}{ME^2} = \frac{1}{ME^2}$

▷ You don't need pilot sample

▷ 1/2 maximizes the standard error hence once we extract our sample, ME cannot be larger (it is equal or smaller)

Overview so far

So far to take decisions: **Control Limits**, **Confidence Intervals**, and **Margin of Error**

Control Limits	Confidence Intervals	Margin of Error
We know the population parameters. We set control limits as a decision rule to determine when a sample statistic that we observe is "out of control," or suggests that the process of interest has likely changed.	We <i>do not</i> know the population parameters. We calculate a confidence interval to estimate a range of plausible values for a population parameter based on the sample.	We <i>do not</i> know the population parameters. We calculate a "back-of-the-envelope" 95% confidence interval by plugging in 2 instead of $z_{\alpha/2}$ or $t_{\alpha/2, n-1}$.

Ch. 15: Exercise 40 (modified)

A city builds a new public parking garage in the central business district. It plans to pay for the structure through parking fees. During a 44-day period, the daily fees collected average \$1,264 with a standard deviation of \$150. Assume that the sample size conditions hold.

(a) Calculate a 95% CI for mean daily income from parking fees.

$$\alpha = 0.05 \Leftrightarrow z_{\alpha/2} = 1.96$$

$$SE(\bar{X}) = \frac{s}{\sqrt{n}} = \frac{150}{\sqrt{44}}$$

$$1,264 \pm 1.96 * \frac{150}{\sqrt{44}}$$

$$[1,219.7 \text{ to } 1,308.3]$$

Ch. 15: Exercise 40 (modified)

A city builds a new public parking garage in the central business district. It plans to pay for the structure through parking fees. During a 44-day period, the daily fees collected average \$1,264 with a standard deviation of \$150. Assume that the sample size conditions hold.

(b) Would this interval be smaller or larger than a 90% CI? Do not actually calculate the 90% CI.

The 95% CI would be larger than a 90% CI. Indeed, the critical value for $\alpha = 0.10$ is $z_{\alpha/2} = 1.64$

(c) How could city planners achieve a narrower 95% CI?

Increase the sample size.

Ch. 15: Exercise 40 (modified)

(d) How many days' worth of data must planners collect to have 95% confidence of estimating the true mean plus \$10 or minus \$10? Does this seem like a reasonable objective?

From class we know, $ME = 2 \frac{s}{\sqrt{n}}$ but don't learn this by heart!! Let's think about it.

The city planners must have a large enough sample so that the width of the CI is \$20. This condition implies that the difference between the upper bound and the mean (i.e., the length of one side of the CI) is \$10.

$$10 = 1.96 * \frac{150}{\sqrt{n}}$$

$$\sqrt{n} = 29.4$$

$$n = 864.36 \approx 865 \text{ days}$$

This objective is not reasonable, as it would take almost three years to complete.

Questions?

Next Deadlines

- Knowledge Check 2 - due Thu 10/1
- Team Assignment on Polls - due Thu 10/1
- Usual weekly OHs (Reed's or Silvia's)
- See you in Berkeley for Block 2

Thank you!

Have a nice evening 😊